# SE456 PROJECT-1: GREENLOGIX SUSTAINABLE ROUTING

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## EXECUTIVE SUMMARY

The GreenLogix Sustainable Routing project can be defined as an engineering solution project that explores how to optimize delivery routes by balancing operational efficiency, economic cost, environmental sustainability, and legal constraints. The aim of the project is not just to be a software idea, but to be seen as a technical product designed to solve a practical optimization problem under regulatory and sustainability conditions.

Route optimization has long been known as one of the most important challenges in logistics and supply chain management. However, today, route determination is no longer limited to simply identifying the shortest path between locations. Factors such as customer demand, delivery windows, travel time, fuel consumption, vehicle capacity, traffic congestion, and environmental impact directly or indirectly influence this optimization.

This problem has become even more visible, especially with the rapid growth of e-commerce and delivery operations after the pandemic [1]. Recent studies in the field of logistics and vehicle routing continue to show that inefficient route planning contributes to failed deliveries, traffic congestion, increased fuel consumption, air pollution and greenhouse gas emissions [2] [3]. For this reason, developing smarter and more adaptable delivery strategies to reduce failed deliveries and optimize operations is extremely important in solving urban logistics problems [4].

The GreenLogix project aims to provide a sustainability-focused routing solution that bridges the gap between advanced optimization methods and real-world delivery constraints. By improving delivery performance and reducing unnecessary travel distance, it can support more environmentally responsible transportation decisions. In this regard, the project aligns naturally with the United Nations Sustainable Development Goals, particularly SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action). Overall, GreenLogix is positioned not only as an optimization tool, but also as a comprehensive software product that addresses long-standing logistics challenges under modern urban, digital, and sustainability conditions.

## STEP 1: Problem Formulation & Research (The "Why")

### 1.1. Literature Review: Vehicle Routing Problem (VRP)

The Vehicle Routing Problem (VRP) is one of the main challenges in the field of combinatorial optimization and integer programming problems that focuses on deciding efficient delivery paths for supplying goods to a set of customer locations [5] [6]. It seeks answers to a set of core questions: vehicle assignment and customer ordering. Furthermore, it determines how much is delivered to each customer, what commodity is delivered, and the precise timing of when each route commences [5]. The historical significance of VRP is closely related to the advancement of modern logistics. It was first introduced by Dantzig and Ramser in 1959, and serves as a generalization of both the Travelling Salesman Problem (TSP) and the Bin Packing Problem (BPP) [6]. A major early milestone in the field was

Clarke and Wright saving algorithm for solving VRP. As one of the earliest practical heuristic methods, it aimed to iteratively improve route sets and efficiently solve relatively large-scale route determination problems. [7]. Over time, numerous algorithms or mathematical models have been developed to reflect the complexity of VRP.

In the literature, there are various variants of VRP that have been studied for years. One of the best-known variants of VRP is the Capacitated Vehicle Routing Problem (CVRP). The basic principle behind CVRP is to find the route that minimizes total delivery costs, without exceeding vehicle capacity limits, and by visiting each customer precisely once [8]. Another major variant is the Vehicle Routing Problem with Time Windows (VRPTW). The basic logic in VRPTW is to identify the optimal set of routes that reduce total operational cost or distance, serving a customer group from a central depot under the condition that every customer receives a single visit without violating capacity limits [9]. In more complex settings, multi-objective models such as the Multi-Objective Capacitated Vehicle Routing Problem (MOCVRP) and the Multi-Objective VRP with Time Windows (MOVRPTW) focus on balancing several conflicting criteria, including total distance, travel time, service quality, and vehicle capacity constraints [10]. On the other hand, environmental issues have led to the emergence of green VRP models such as the Fuel Consumption Vehicle Routing Problem (FCVRP) and the Green Capacitated Vehicle Routing Problem (GCVRP), which seek to reduce both travel distance, fuel consumption and CO2 emissions in route planning [10], [11].These variants are necessary because real delivery systems must take into account factors such as limited vehicle capacity, customer-specific service intervals, varying vehicle capabilities, warehouse structures, and variable travel conditions.

From a computational perspective, VRP remains important because obtaining effective solutions for real-world routing scenarios is demanding. As the number of customers or vehicles increases, a problem arises where achieving the optimal result becomes more difficult, known as NP-hard [5], [10]. Therefore, the literature generally distinguishes between two primary strategies: 'exact methods' that guarantee the optimal solution but are often too slow for large datasets, and 'heuristic/metaheuristic methods' that enable near-optimal and feasible solutions in a short time [12]. In VRP, for example, the goal is not to find the perfect mathematical result; it is to have a feasible and logical route within a limited time. The main value of routing software related to managing the computational complexity efficiently [5], [10].

One of the main reasons why VRP remains relevant is that logistics operations have become increasingly dynamic and uncertain. In real-world scenarios, traffic conditions, customer demand, or order arrival times do not generally remain constant; thus, static routing plans may be insufficient [13]. Many studies in the literature show that routing is not just about determining where vehicles will go, but also about when vehicles will depart and how this timing will be compatible with inventory management and delivery times [14] . For this reason, routing has transformed from a simple route-finding problem into a time-dependent and decision-making problem [15].

The structural complexity, constraints, and computational challanges of VRP drive the development of various optimization methods. Among these, metaheuristic algorithms such as Genetic Algorithms and Simulated Annealing are widely utilized for providing high-quality solutions.

### 1.2. Comparison of Algorithms

The structural complexity, constraints, and computational challenges of VRP drive the development of various solution methods. These methods can be categorized into heuristic and exact algorithms. Exact algorithms include some approaches such as branch and bound, cutting planes, network flow and dynamic programming approach, whereas heuristic consist of contruct, improved and sub-heuristic methods[16]. Among these, sub- heuristic algorithms such as Genetic Algorithms, Simulated Annealing, Neural networks, Particle Swarm Optimization and Ant colony algorithms are widely utilized for providing high-quality solutions [16]. In the recent years, hybrid algorithms combining two or more algorithms have become popular. Notable instances of hybrid algorithms involve iterative methods with integer programming, Machine Learning (ML) with Tabu Search (TS), Local Search (LS) with both Genetic Algorithm (GA) and Ant Colony Optimization (ACO) [17].

**Genetic algorithms(GA)** are well-known metaheuristic methods inspired by the principles of Darwinian evolution. The process is iterative, each iteration continues until a stopping condition is carried out such as maximum number of generations or a target solution quality is satisfied[18]- [19]. For example, in a study, each offspring(or individual) is to minimize total distance by utilizing a route elimination and local search. Although GA approaches are still used and support product-oriented idea, Hybrid Generic algorithms (HGA) can be suitable sometimes when product must balance several routing aims simultaneously such as distance, vehicle count, time-window feasibility, and potentially sustainability-related metrics [20]. Moreover, GA can be efficient at searching a broad area but it struggles to find optimal solution in large datasets due to getting stuck in local optima (false peaks). To addresss this, hybrid methods combining GA and Tabu Search can find shorter and more efficient routes by leveraging GA’s exploration power and Tabu Search’s local refineminet [21].

**Simulated Annealing (SA) imitate**s metallurgical cooling by using a "temperature" parameter to move away from local minima; at higher temperatures, the algorithm offers inferior solutions to broaden the search. A variant, Deterministic Annealing enhances VRP results by replacing random number generationwith deterministic threshold calculations [18] . The Hybid GA-SA outperforms GA and SA alone by using the power of population based searching. Standart SA can only propose single solution at a time which is extremely slow and prone to miss better areas of the search space, whereas hybrid approaches can integrate GA’s population based global search with SA’s stochastic local refinement. Therefore it can explore multiple regions of the search space simultaneously by yielding shorter and more optimized routes in a short time [22] .

**Deep Reinforcement Learning (DRL)** has emerged as a novel approach in VRP researches. Unlike traditional methods, DLR can generate end-to-end routing policies or improve solutions step-by-step through interaction with dynamic environments by leveraging encoder-decoder structures and atetntion mechanisms. However, despite its adaptibility and high

speed, it can struggke to accomplish precise optimizetion levels for large datasets [23]-[24]. Thus, hybrid approaches that combines DRL with Local search algorithms can provide superior balance integrating learning based flexibility with mathametical correnctness to find shorter and effective routes [25].

In summary, while Genetic Algorithms enable powerful exploration capabilities, Simulated Annealing offers a simpler local search mechanism, and Deep Reinforcement Learning offers a more adaptable and learning-based approach for dynamic routing environments. However, hybrid approaches seem to be a better option because of their exploration and solution quality. This makes hybrid approaches more suitable for practical VRP.

| Method | Core Mechanism | Strengths (Advantages) | Limitations (Disadvantages) | Hybrid Advantage (The "Why") |
|---|---|---|---|---|
| **Genetic Algorithm (GA)** | Mimics Darwinian evolution (Selection, Crossover, Mutation). | Good at exploring a broad search space (Global Search). | Prone to getting stuck in local optima (false peaks) in large datasets. | GA + Tabu Search: Combines GA's exploration with Tabu Search's local refinement to escape local traps. |
| **Simulated Annealing (SA)** | Mimics metallurgical cooling using a "temperature" parameter. | High capability to escape local minima by accepting inferior solutions early on. | Extremely slow as it processes a single solution at a time; limited global search. | Hybrid GA-SA: Integrates GA's population-based search with SA's stochastic refinement for faster, high-quality results. |
| **Deep Reinforcement Learning (DRL)** | Interaction-based learning via Encoder-Decoder & Attention mechanisms. | Exceptionally fast and highly adaptive to dynamic, real-time environments. | Struggles with mathematical precision and optimality levels for very large-scale datasets. | DRL + Local Search: Balances learning-based flexibility with mathematical correctness to ensure shorter, effective routes. |
| **Hybrid Genetic Algorithm (HGA)** | Multi-objective optimization (Product-oriented). | Successfully balances distance, vehicle count, time windows, and sustainability. | Requires more complex parameter tuning and higher computational setup. | Ideal for complex VRPs where several routing aims must be satisfied simultaneously. |

### 1.3. SDG Mapping and Sustainability Relevance

GreenLogix is naturally aligned with Sustainable Development Goals 11 and 13. According to the official UN Sustainable Development Goals website, Goal 11 aims to "make cities and human settlements comprising, reliable, robust and sustainable," while Goal 13 assesses urgent calls for action to fight climate change and its effects. GreenLogix Sustainable Routing can be directly linked to **SDG 11 (Sustainable Cities and Communities)** because of its contribution to more efficient and less disruptive urban logistics operations. While SDG 11.2 primarily focuses on safe and accessible transportation systems, GreenLogix tries a meaningful contribution to this by preventing unnecessary vehicle circuits and increasing route efficiency for urban freight transport. It may provide operational enhancement, particularly in large cities with high traffic density, by supporting more planned movement of delivery fleets. Our product also focuses on the 11.6 target, which aims to minimize negative environmental influence in cities. GreenLogix may improve delivery processes by optimizing unnecessary distances, waiting times, and fuel-intensive deviations. Thus, rather than a general sustainability claim; GreenLogix offers a real-world operational contribution to solving problems due to reducing vehicle mileage, lowering traffic inefficiency, and preventing the burden of urban air pollution [26].

On the other hand, GreenLogix also directly linked to SDG 13.2, which predicts the integration of climate change measures into organizational policies and planning processes. It aims to effective logistics planning by treating fuel consumption and estimated CO2 emissions not only as a secondary alternative, but as a main component of the route-determining stages. Thus, route selections are optimized not only according to distance and time parameters, but also considering environmental costs. Furthermore, GreenLogix personalizes climate action (SDG 13). By providing clear data on fuel and emissions for each route, we aim to give teams a broader perspective. This can help managers with the tools they need to lead sustainably and also provide daily operational data for emission reductions. It can also support more informed decisions and enable measurable emission-reduction strategies in day-to-day operations [27]

### 1.4. Legal and Ethical Analysis

The legal and ethical responsibilities of the GreenLogix project primarily comprise driver-related personal data such as identity information, work schedules, route history, live location data, and vehicle-assignment records. According to Turkish Data Protection Law (KVKK), the data controller must clearly inform the data subject about the identity of the controller, the aim and legal situation of processing, data transfers and the rights of the individual. For GreenLogix, this process involves limiting data collection activities to only those that are "truly necessary" for route optimization and security, thus adopting a design that comprises to the principles of data minimization and purpose limitation [28].

A second issue concerns automated decision-making. Under GDPR Article 22, individuals are protected against decisions according to automated processing only if those decisions have legal or similarly important outcomes. In the context of GreenLogix, this becomes crucial if automated route suggestion significantly influence driver workload, break opportunities, safety risks, or performance evaluation. Therefore, the system should be treated as as a human-supervised decision-support tool rather than a fully autonomous route-determining authority [29].

Liability and ethical issues are also a crucial part of these legal frameworks. If an automated route suggestion leads to an accident or a violation of rest stop rules, liability can lead to complex legal processes including the software provider, the management company, the authorized person approving the route. In this context, the legal distinction between an algorithmic recommendation and a mandatory instruction is vital for determining fault in regulatory violations.Thus, the safest way to prevent this uncertainty is to design GreenLogix as a "human-supervised decision support system" rather than an "autonomous decision-maker”. To minimize legal risks, the software must maintain transparent audit records to prove it does not violate security protocols or driver rights. Furthermore, from an ethical perspective, efficiency and fuel economy goals should not compromise driver safety and fair working conditions. Consequently, the protection of personal data, measured data use, and operationalclarity should be considered not only legal safeguards but also main product requirements for GreenLogix.

## STEP 2: Technical Product Specification (The "What")

This section explains the mathematical fomulation, implementation tools, and operational constraints used in the GreenLogix prototype. First, an evaluation engine was created, and then a lightweight optimization layer was constructed. The goal is to create a sustainability-conscious route planning system by balancing logistics efficiency with environmental influence and operational applicability.

### 2.1. Functional and Nonfunctional Requirements

### Functional Requirements

- FR-1 :The system will allow users to add, edit, and remove delivery stops through an interactive input interface.
- FR-2 : The system will convert text-based addresses into geographic coordinates by using an external geocoding service.
- FR-3 :The system will produce an optimized stop order using a weighted evaluation function based on distance, fuel consumption, emissions, and operational penalties.
- FR-4: The system will adjust fuel consumption and $CO_2$ emmission according to the remaining cargo load on each route segment.
- FR-5: The system will compare the delivery request with the vehicle capacity and will mark the route as unsuitable if the capacity limits are exceeded.
- FR-6: The system will monitor continuous driving duration and apply penalties when predefined driving limits are exceeded.
- FR-7: The system will obtain and show actual road-based route geometry from an external routing service instead of relying only on straight-line visualization.
- FR-8: The system will provide an interactive map interface showing depot locations, stops, and route paths.
- FR-9: The system will produce a step-by-step driver operations log showing delivery actions and remaining cargo throughout the route.
- FR-10: The system will preserve user-entered stop data and optimization outputs across interface interactions within the active session.

**Non-Functional Requirements**

- NFR-1:The system should provide a clear and simple dashboard interface that allows logistics users to define a scenario and generate route results with minimal interaction.
- NFR-2: The system should return optimization results within an acceptable response time for small-scale delivery scenarios.
- NFR-3: The system should provide stable and internally consistent route distance and sustainability estimates suitable for prototype-level evaluation.
- NFR-4:The modular architecture should support future extension toward larger routing scenarios and multi-vehicle optimization with limited redesign of core components.
- NFR-5: The system should handle missing, invalid, or incomplete user inputs gracefully without causing application failure.
- NFR-6: The application should be accessible through a Streamlit-based dashboard interface.
- NFR-7: The system should use a defined $CO_2$ emission factor for diesel-based fuel estimation to support sustainability-oriented route analysis.
- NFR-8: The Python-based implementation should be portable across common operating systems through a standard virtual environment setup.
- NFR-9: The service-oriented and modular design should allow future replacement or upgrade of routing and geocoding components.
- NFR-10: The dashboard should present key route metrics, route status, and user feedback in a visually clear and easily interpretable format

## 2.2. Mathematical Foundation

GreenLogix is designed as a route evaluation and decision support system that provide economic cost and environmental sustainability. In this current implementation, the system evaluates candidate delivery routes and selects improved route orders by combining distance, fuel consumption, carbon emissions, travel time, and operational penalties within a unified scoring logic. Rather than treating routing as a pure shortest-path problem, the prototype assess route quality as a multi-factor engineering decision problem [30]-[31]-[32]:

The route be represented as an ordered set of stops:

$$\mathbf{R = [s_0, s_1, s_2, \ldots, s_n, s_0]}$$

where: $s_o$: depot / remanings: customer locations visited in order before returning to the depot

The total route distance is measured as:

$$\boldsymbol{D_{total} = \sum_{i=1}^{n-1} d(s_i, s_{i+1})}$$

where, d(s_i,s_(i+1)): the network-based distance between two consecutive stops. Likewise, the total travel time is measured as:

$$\mathbf{T_total = \Sigma\, t(s_i, s_{i+1}) + \Sigma\, service_time_i + break_penalties}$$

where $t(s_i, s_{i+1})$ : segment travel time/ break penalties and service time also affect whole route assesment.

In the current GreenLogix prototype, both distance and time are taken from an external routing service, rather than from a separate internal traffic model.

Fuel consumption is measured utilizing cargo load carried by the vehicle at that point in the route. Segment-level fuel consumption is modeled as

$$\mathbf{F_i = base_fuel_rate \times d(s_i, s_{i+1}) \times (1 + \alpha \times load_i) \times traffic_factor_i}$$

**where base_fuel_rate**: vehicle's baseline fuel usage per kilometer / **α:** load sensitivity coefficient / **load$_i$**: remaining cargo before traveling the segment / **traffic_factor$_i$** : traffic-related variation

Total fuel consumption:

$$\mathbf{F_total = \Sigma\, F_i}$$

Carbon emissions are then calculated from total fuel consumption utilizing emission coefficient:

$$\mathbf{E_CO_2 = F_total \times \beta}$$

where β: the emission factor associated with the selected fuel type.

The route assessment also includes operational feasibility checks. Penalties may be added when route conditions violate some limits such as vehicle capacity, fuel availability, continuous driving duration, or late arrival conditions. Therefore, the final route score is calculated not based on distance alone, but also on a weighted combination of logistics and sustainability criteria:

$$\textbf{Score} = w_1 \cdot D_{total} + w_2 \cdot F_{total} + w_3 \cdot E_{CO_2} + w_4 \cdot \text{Penalty}$$

### 2.3. Tool Selection

GreenLogix uses a simple and modular software tool. The main programming language is Python because it supports rapid prototyping, clear mathematical implementation, and easy testing. “The dashboard layer is implemented in Python/Streamlit and integrated with modular routing, fuel, and constraint services. The external service logic is divided into two service modules: geocoding_service.py for converting addresses to geocoordinates, and routing_service.py for retrieving route-based geometry, distance, and estimated travel time [33]-[34]-[35]-[36]-[37]-[38].

| Tool | Role | Justification |
|---|---|---|
| Python | Core implementation language | It is easy to implement, test, and explain |
| Streamlit | Dashboard interface | It enables rapid development of an interactive, product-oriented prototype. |
| Folium | Interactive route visualization | It allows the route, depot, and delivery stops to show in an interactive map |
| Geocoding API service | Address-to-coordinate conversion | This allows for the creation of realistic routes from textual addresses. |
| Routing API service | Route generation and travel estimation | It provides realistic route geometry, distance, and duratio |
| Plain Python service modules | Modular architecture | They separate geocoding and routing logic from the main Streamlit application |

### 2.4. Constraint Design

The current GreenLogix may show realistic routing conditionsin instead of arguing that the shortest path is always the best solution. For this reason, the prototype incorporates several practical operational constraints into its route evaluation logic [39]-[40]-[41].

**Vehicle capacity :** routes are checked against total delivery demand. If required cargo exceed maximum capacity, route can be marked as infeasible and penalties can be applied

**Fuel availability:** system evaluates whether vehicle can complete route within gits fuel limit.

**Continous driving duration:** system monitors how long vehicles operates withoud a break or applies penality when driving conditions are violated.

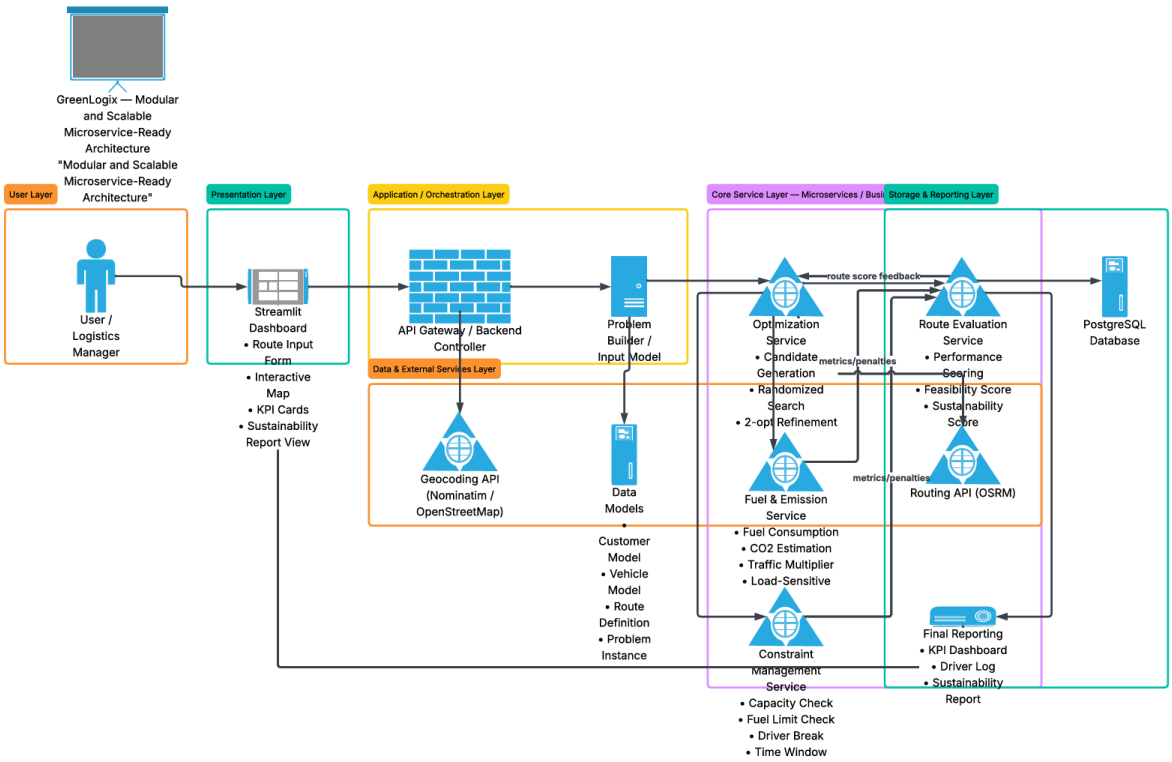
**Load sensitive fuel behaviour and traffic multipliers:** not difficultconstraint but they can influce route quality

There are also some other constraints for future development but are not yet fully implemented in the current prototype: heduling logic, richer traffic inputs, and more advanced optimization strategies. Finally, while the current GreenLogix system includes practicalconstraints suchs fuel acalibility and vehicle capacity limits, these will focus more advanced operational constraints for future versions of the product**.**

**STEP-3: Engineering Design and UX**

**3.1 System Architecture**

GreenLogix system utilizes independent modules to improve system maintainability and scalability rather than evaluating the routing process from a single perspective. This also enables teansitiong to service-based or cloud-based deployments in the future. The center of the architecture is the data model layer, that recognizes the system's variables, such as vehicles, route definitions, and problem instances. Moreover, the fuel and emissions module calculates fuel consumption and carbon emissions according to distance, traffic multiplier, and dynamic vehicle load. The constraint evaluation module controls whether a candidate route violates key operational rules such as vehicle capacity, fuel limits, mandatory driver breaks, and delivery time intervals. These outputs are then utilized by the route solver, which assess complete routes and produces an integrated performance score. Finally, the optimization module produces multiple route candidates and improves them using random search and optimization. The Streamlit dashboard esus as the user interface and visualization layer.This ensures that GreenLogix is viewed not just as a prototype interface, but as a structured optimization product. The modular separation between data representation, fuel modeling, constraint validation, route scoring, and optimization enables testing and scalability of the system. For example, the existing application can later be improved by replacing the current system with more advanced hybrid optimization strategies without redesign the whole product. As a result, the system can be treated as a sustainable software pr**oduct.**




### 3.2 Proposed Architecture Components

GreenLogix's architecture is defined by six core components. The first component is the User Interface Layer providing the operational dashboard where the user enters delivery stops, defines vehicle parameters, initiates optimization, and interprets route metrics. The second component is the Input and Data Modeling Layer including customer requests, vehicle settings, and route problem definitions. The third component is the Fuel and Sustainability Engine calculating fuel consumption and CO2 emissions under load-sensitive and traffic-sensitive conditions. The fourth component is the Constraint Management Layer checking route suitability according to capacity, fuel availability, rest stop requirements, and time-related penalties. The fifth component is the Optimization Engine producing route candidate. The sixth component is the Visualization and Reporting Layer presenting outputs such as distance, fuel, cost, emissions, score, and operational records.

### 3.3 Product Backlog: User Stories

1. As a logistics manager, I want to create a highly efficient route to minimize operational costs and fuel consumption.

2. As a fleet operator, I want to define vehicle capacity and fuel parameters to reflect route planning for real-world scenarios.

3. As a sustainability officer, I want to obtain estimated CO2 emissions per route to support environmental decision-making processes.

4. As an operations analyst, I want to compare route score, fuel usage, and travel time to evaluate the  balance between efficiency and sustainability.

5. As a dispatcher, I want the system to alert me unfeasible routes to avoid violating operational rules.

6. As a driver supervisor, I want mandatory driver breaks to ensure routes remain compliant with safety standards.

7. As a planner, I want to automatically test multiple route candidates without not need  to manually evaluation of  all route combinations.

8. As a decision maker, I want to visual summary of routing results on a dashboard to quickly interpret operational performance.

9. As a system user, I want the ability to edit stop requests and route inputs to simulate different delivery scenarios.

10. As a product owner, I want the system to be modular to easily integrate future optimization methods and real map services.

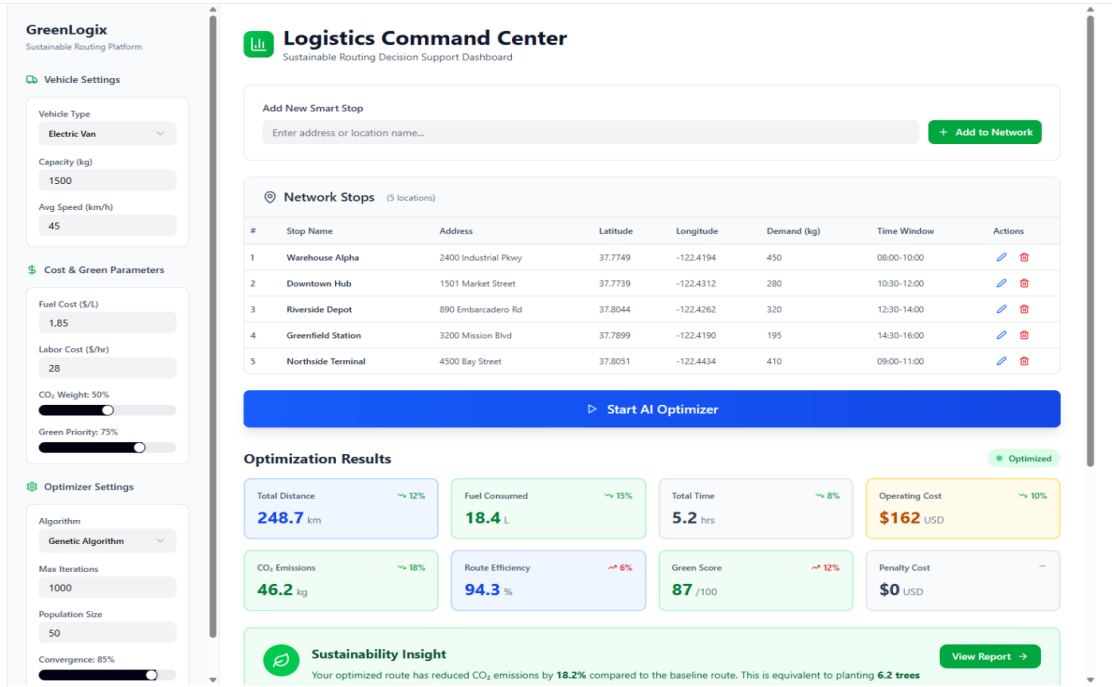
**3.4 High-Fidelity Wireframe / Dashboard Design**

The GreenLogix dashboard is designed as a decision support interface rather than a simple route viewing page. The main aim is to present a more understandable engineering resu for users. The dashboard includes configurable vehicle settings, sustainability parameters, optimization controls, a route creation trigger, a results panel, map-based route visualization, and an operational driver log. The metrics section summarizes the most important route KPIs, including total distance, total fuel, total time, total cost, CO2 emissions, route efficiency, overall score, and penalty value. Thisenables the user to quickly evaluate whether the generated route is practically acceptable and environmentally suitable. The map section visualizes the route spatially, while the driver operation log provides explainability by showing the sequence of route actions and unloading steps. From a user experience perspective, the dashboard groups relevant controls and results in separate panels. Vehicle settings and optimization parameters are located in the sidebar to provide the main screen focuses on route outputs. Therefore, the dashboard is presented not only as a visual front-end but also as an engineering communication layer.

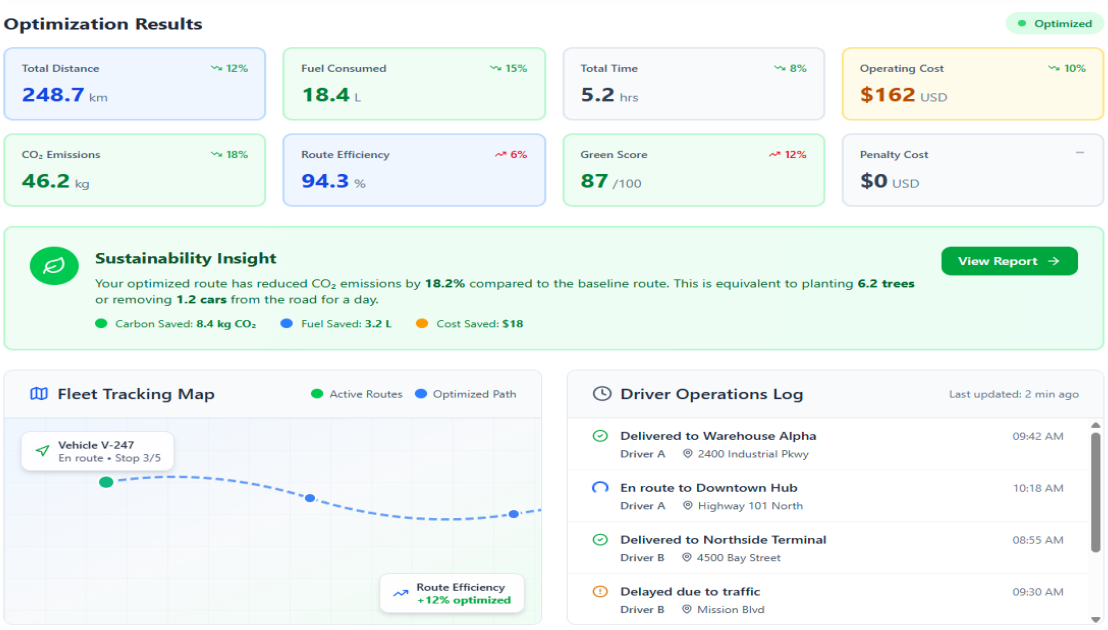
## Scenario 1 - Main Dashboard Overview

The initial dashboard presents the full GreenLogix workflow in a single view. It combines vehicle and sustainability settings, smart-stop input, the network stop table, the AI optimizer trigger, optimization KPIs, the sustainability insight panel, a fleet tracking map, and the driver operations log.

### Dashboard view




*Upper interface region*

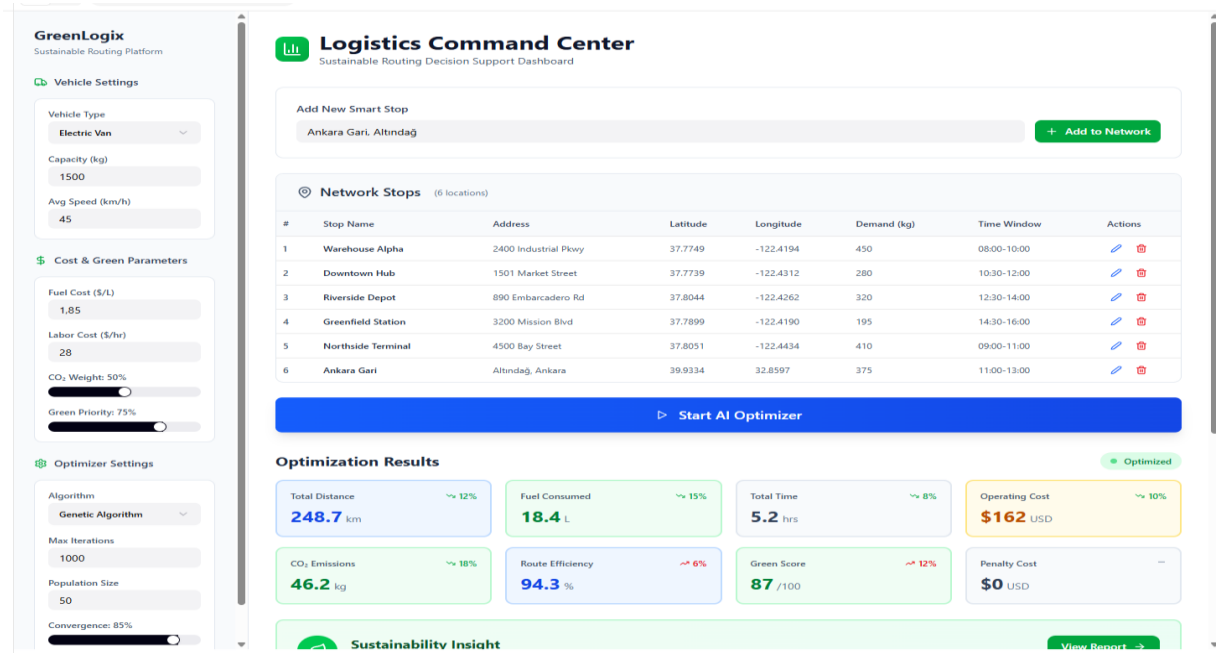



*Lower results, map, and operations region*

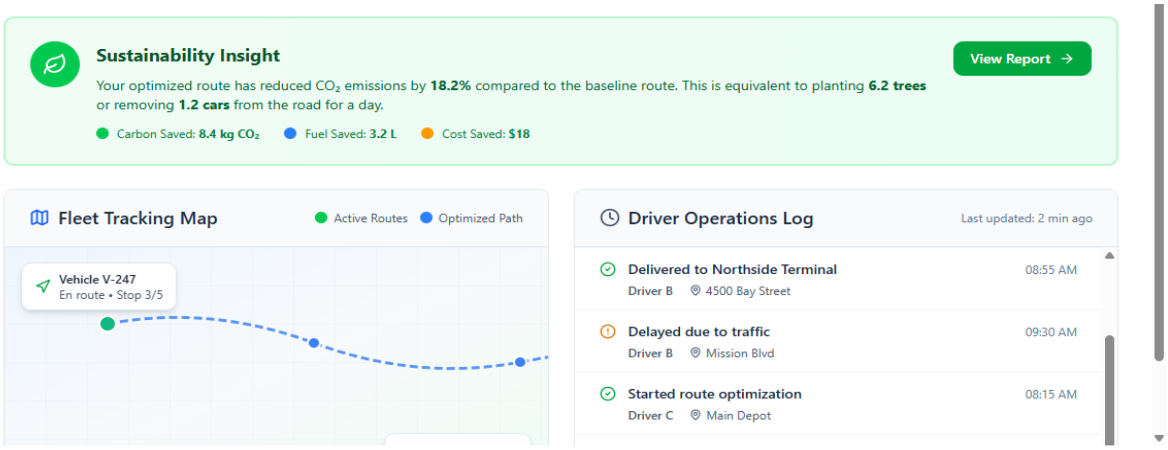
## Scenario 2 - Add New Stop

This scenario shows the stop-entry workflow after a new address has been added. The smart-stop input is populated with 'Ankara Gari, Altindag', and the network table expands to six configured locations, demonstrating that new delivery points can be incorporated without altering the overall dashboard structure.

## Dashboard view




*Upper interface region*

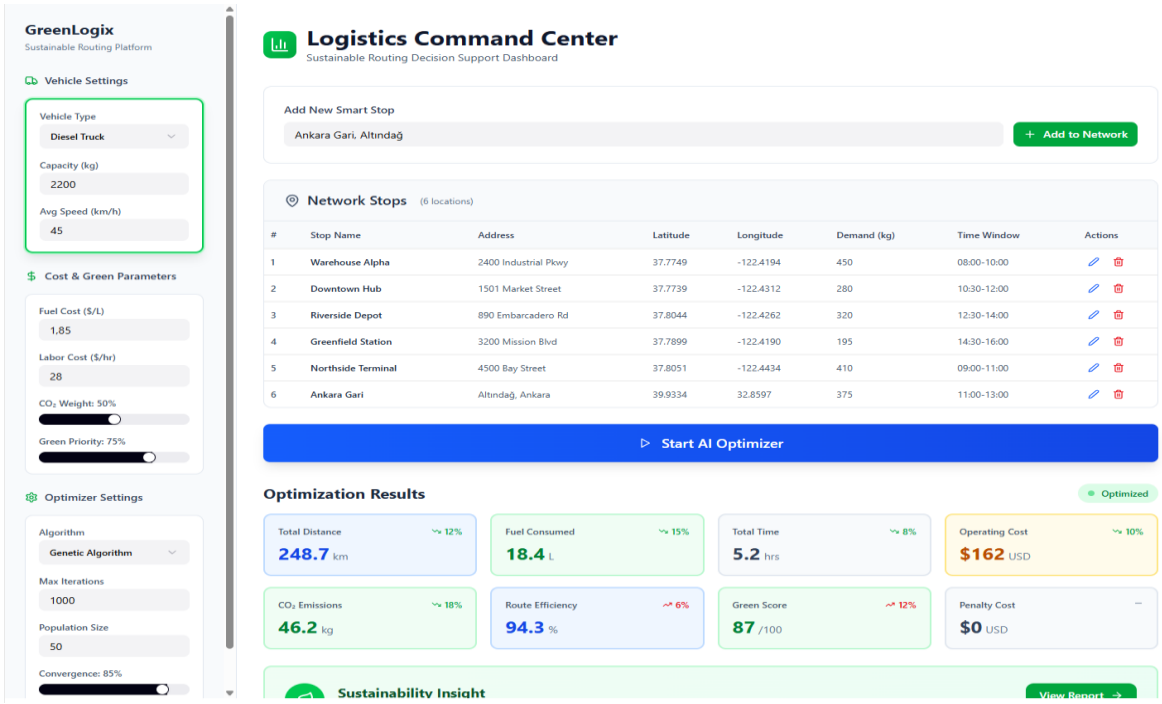



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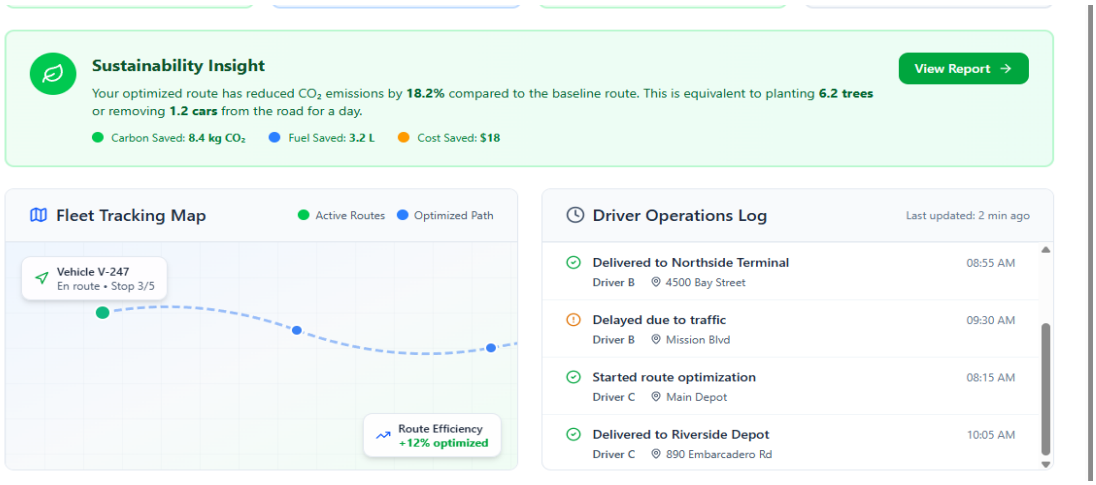
## Scenario 3 - Vehicle Settings Configuration

This screen highlights the Vehicle Settings panel and reflects updated operational inputs. The selected fleet becomes a diesel truck with a 2,200 kg capacity and a 45 km/h average speed, illustrating how transport assumptions can be adjusted before optimization is launched.

## Dashboard view




*Upper interface region*

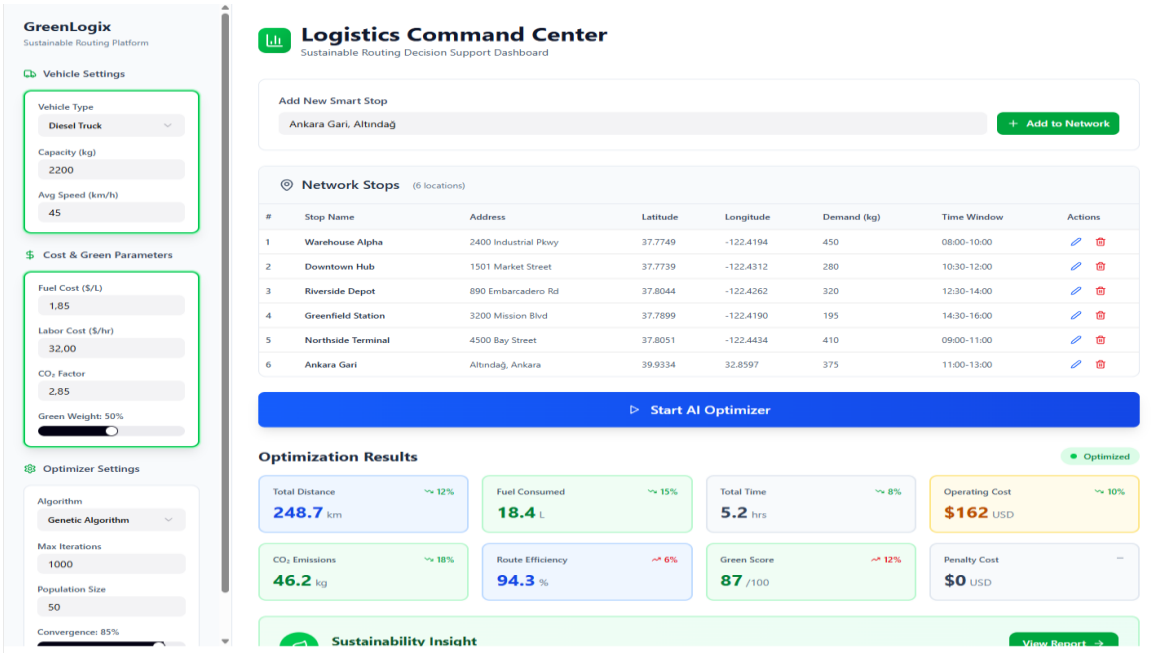



*Lower results, map, and operations region*

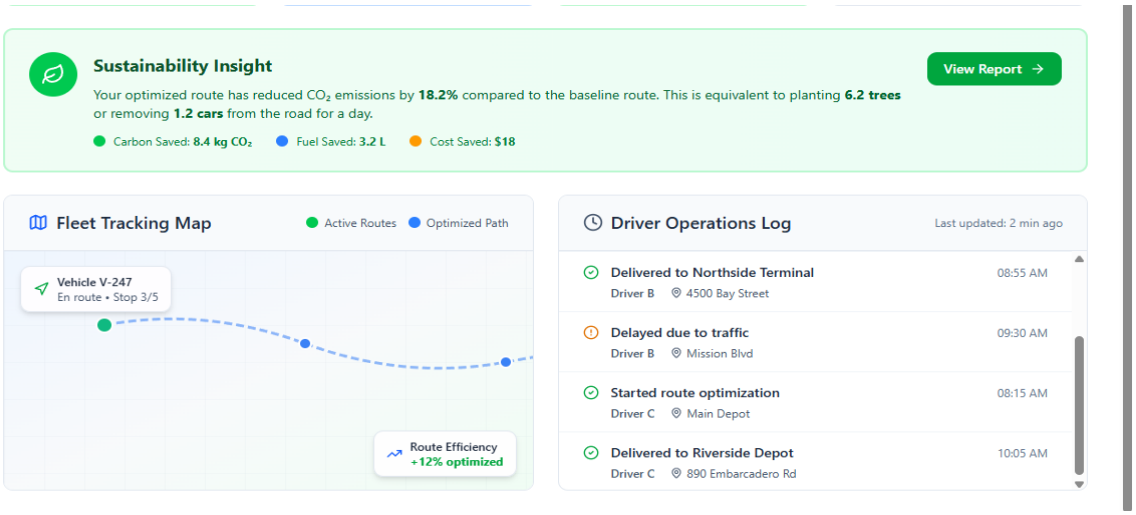
## Scenario 4 - Cost and Green Parameters

This scenario emphasizes the sustainability-cost controls used in multi-criteria route evaluation. Fuel cost, labor cost, the CO2 factor, and green weighting are updated and highlighted to show how environmental and financial priorities can be tuned by the planner.

## Dashboard view




*Upper interface region*

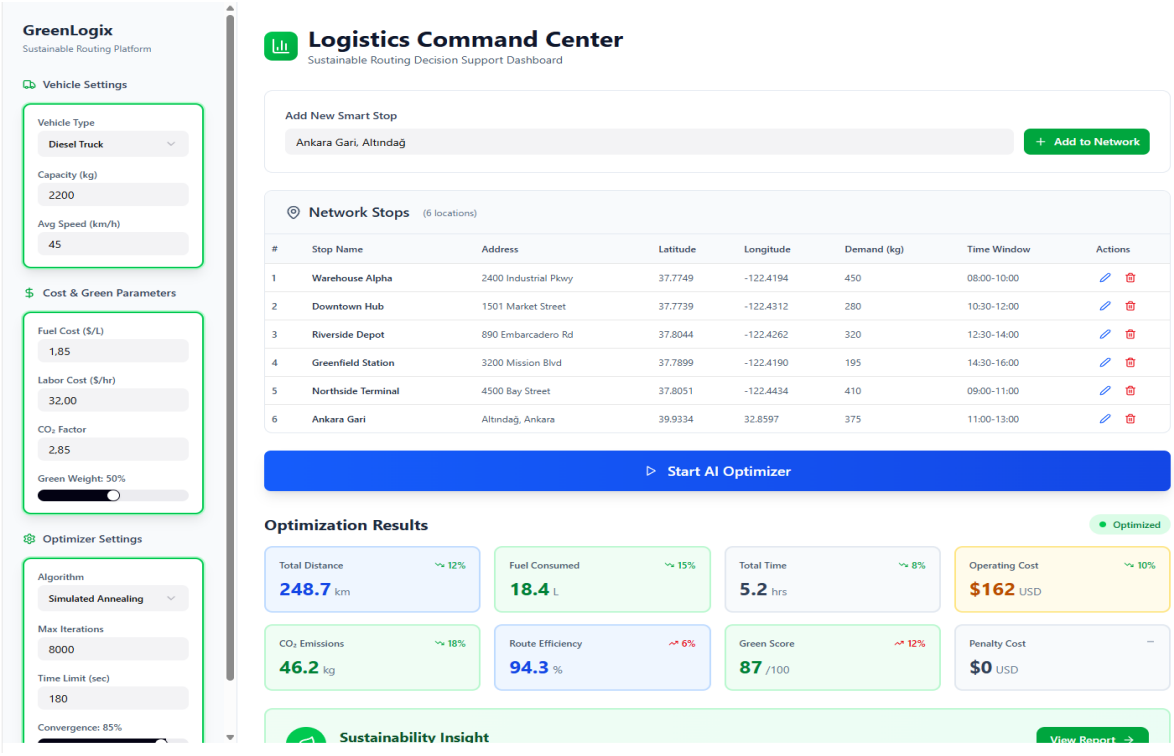



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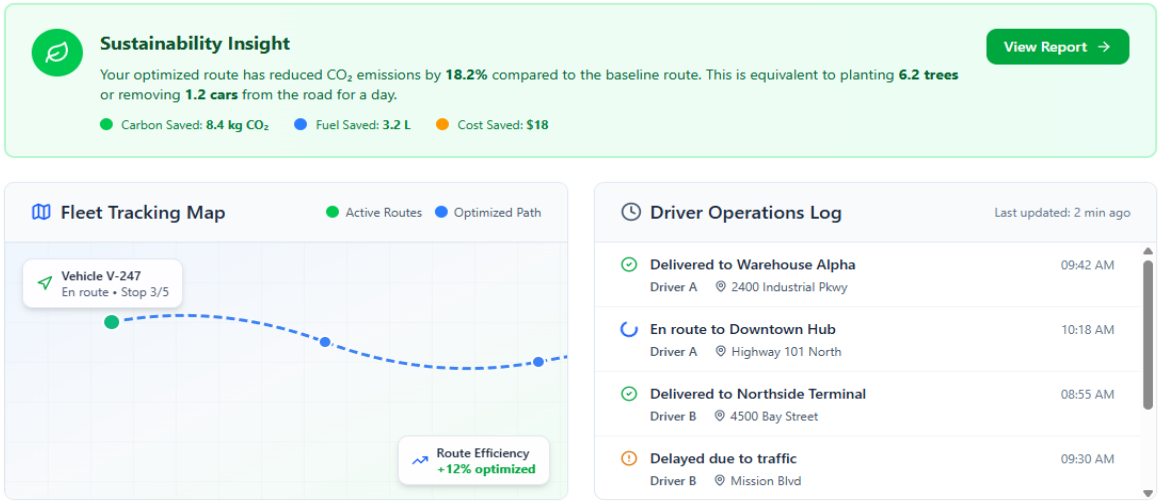
## Scenario 5 - Optimizer Settings

The Optimization Settings panel is foregrounded to show algorithm-level configuration. Simulated Annealing is selected, and the iteration limit and time limit are increased, demonstrating how the search strategy can be refined for deeper optimization.

## Dashboard view




*Upper interface region*

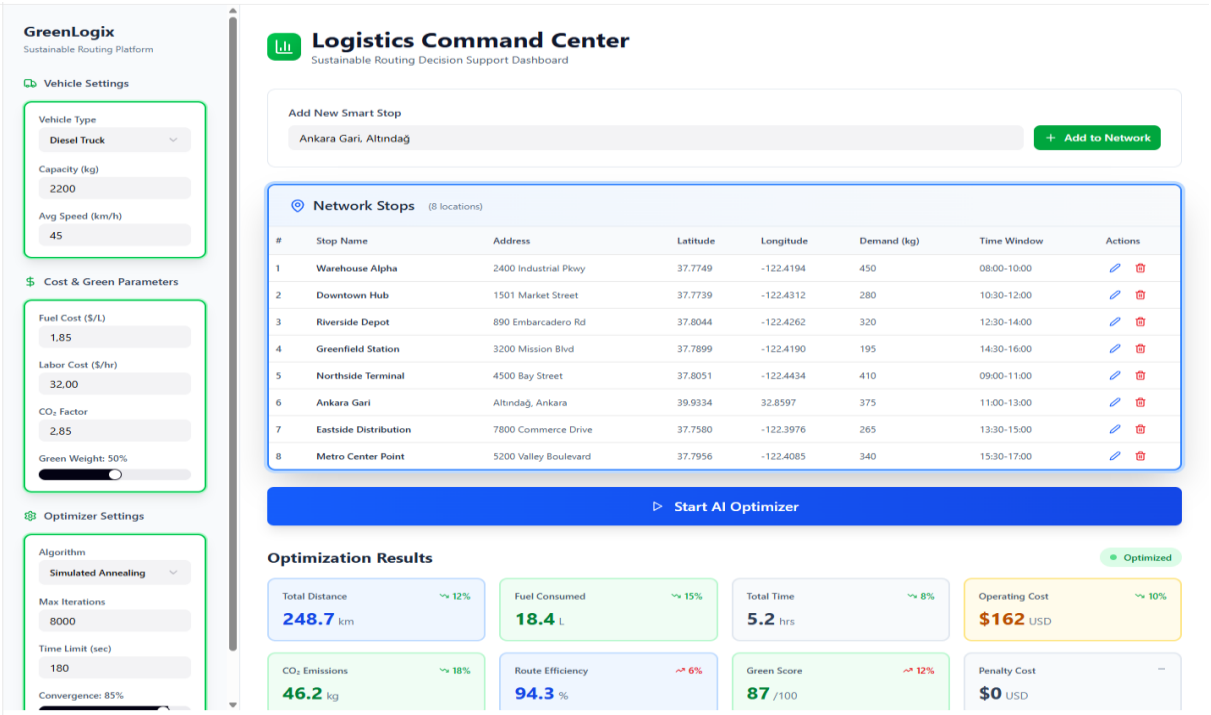



*Lower results, map, and operations region*

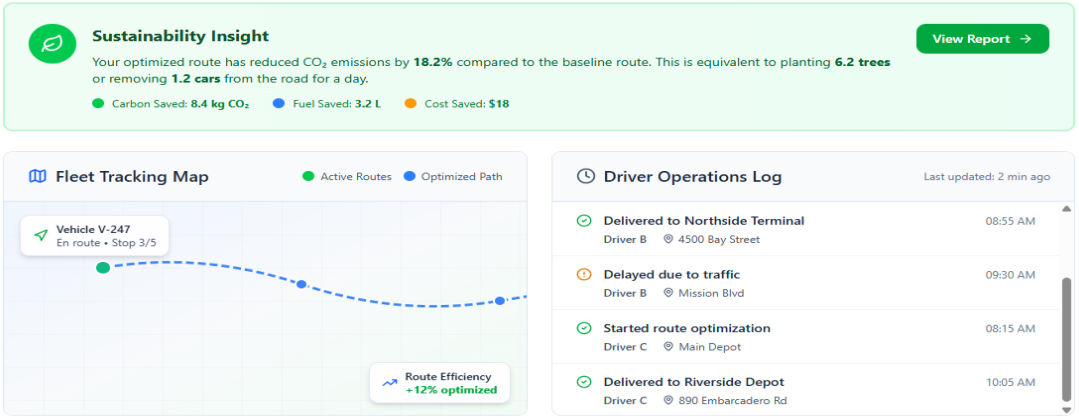
## Scenario 6 - Expanded Network Stops Table

In this scenario the route table becomes the main focal element. Two additional delivery points are added, increasing the total to eight stops and illustrating a denser planning case with a larger routing problem instance.

### Dashboard view




*Upper interface region*

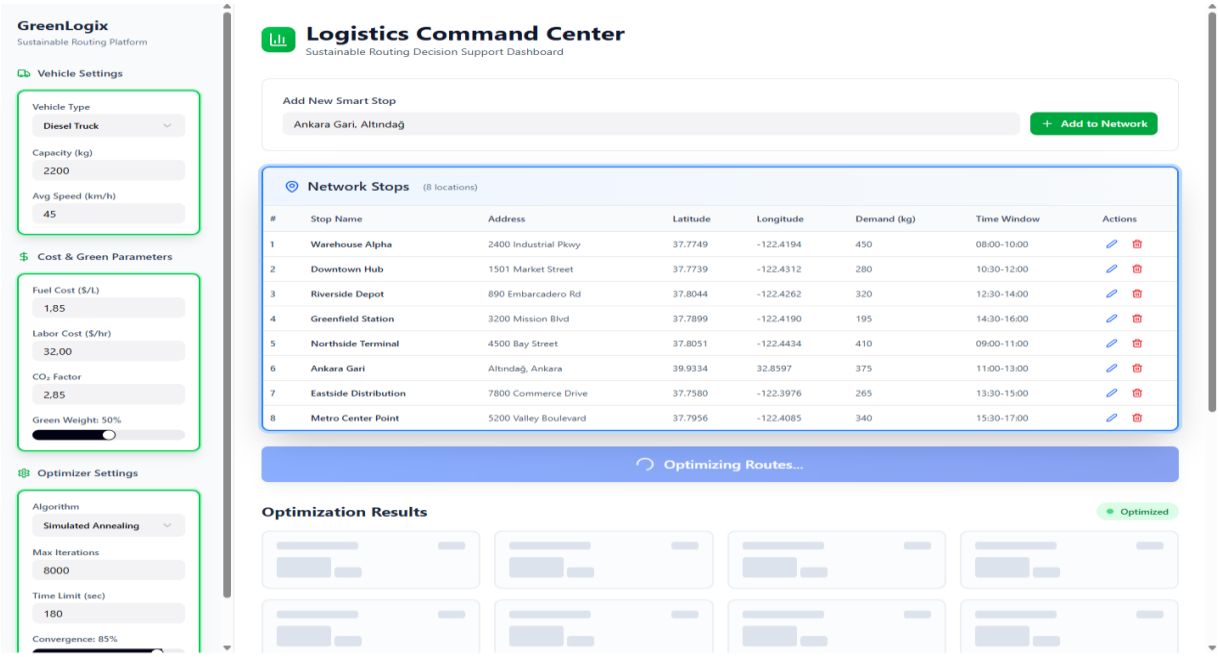



*Lower results, map, and operations region*

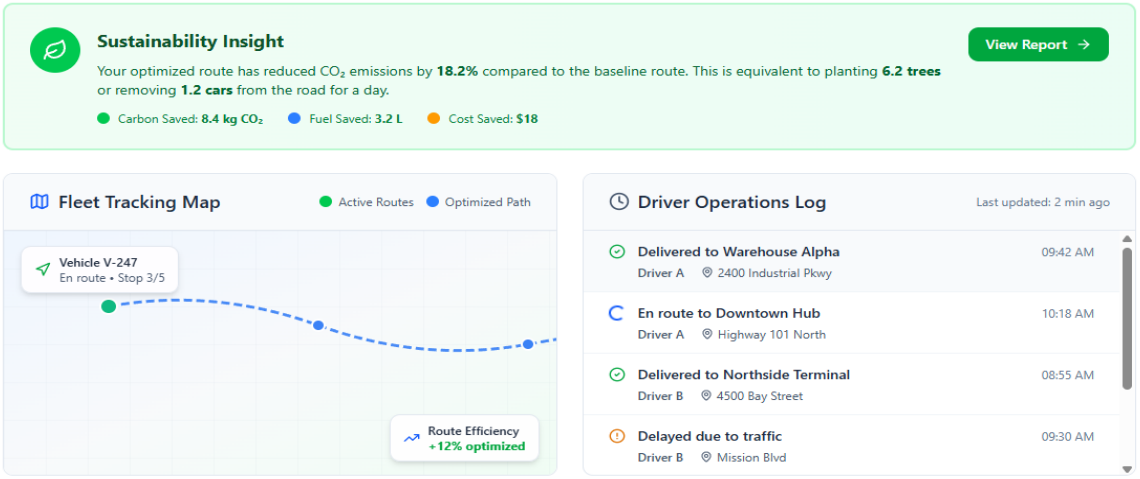
## Scenario 7 - Optimization Running State

This view represents the dashboard while optimization is in progress. The primary action switches to 'Optimizing Routes...' and the KPI cards are replaced by loading placeholders, signaling that the system is recalculating metrics and route quality.

## Dashboard view




*Upper interface region*

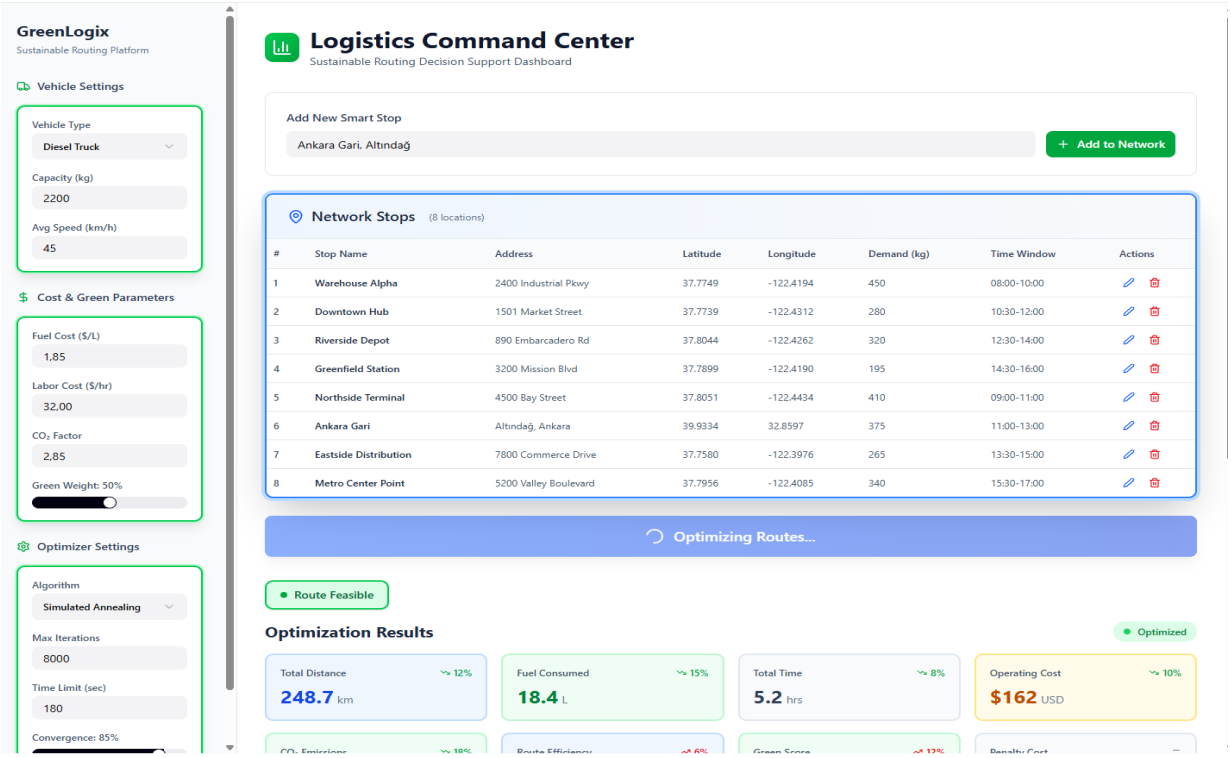



*Lower results, map, and operations region*

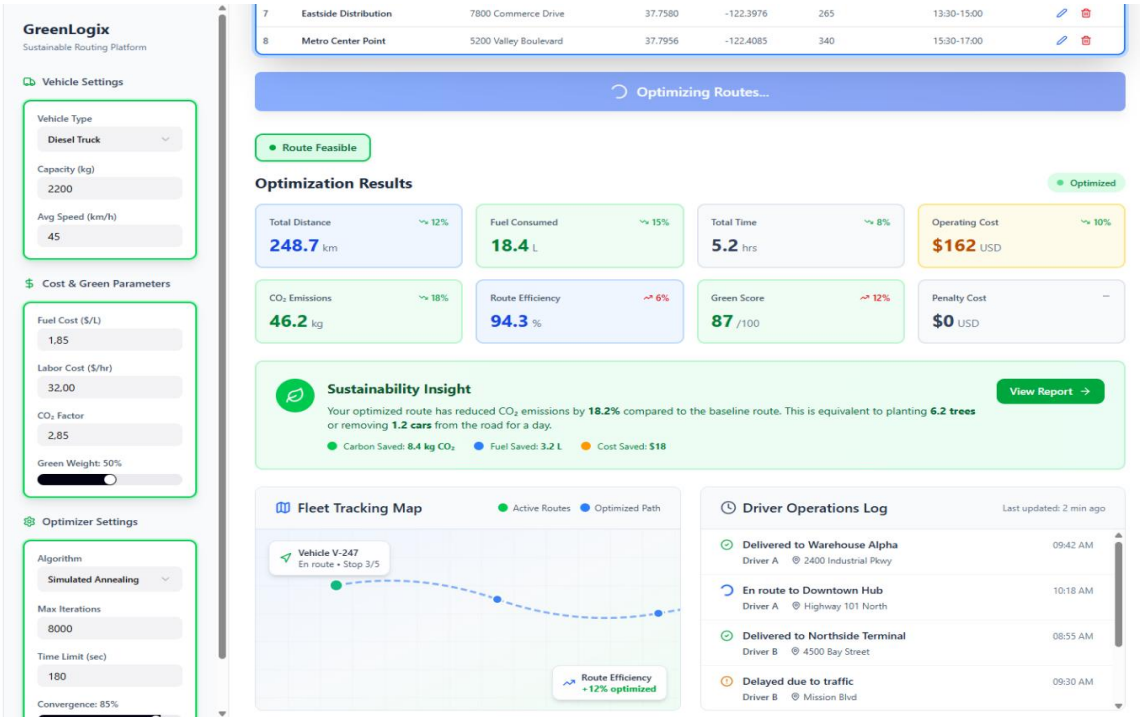
## Scenario 8 - Feasible Route Result

This scenario presents a successful result state. A green 'Route Feasible' status badge appears above the results section, and the KPI cards remain fully populated, communicating that the optimized route satisfies operational constraints.

## Dashboard view




*Upper interface region*

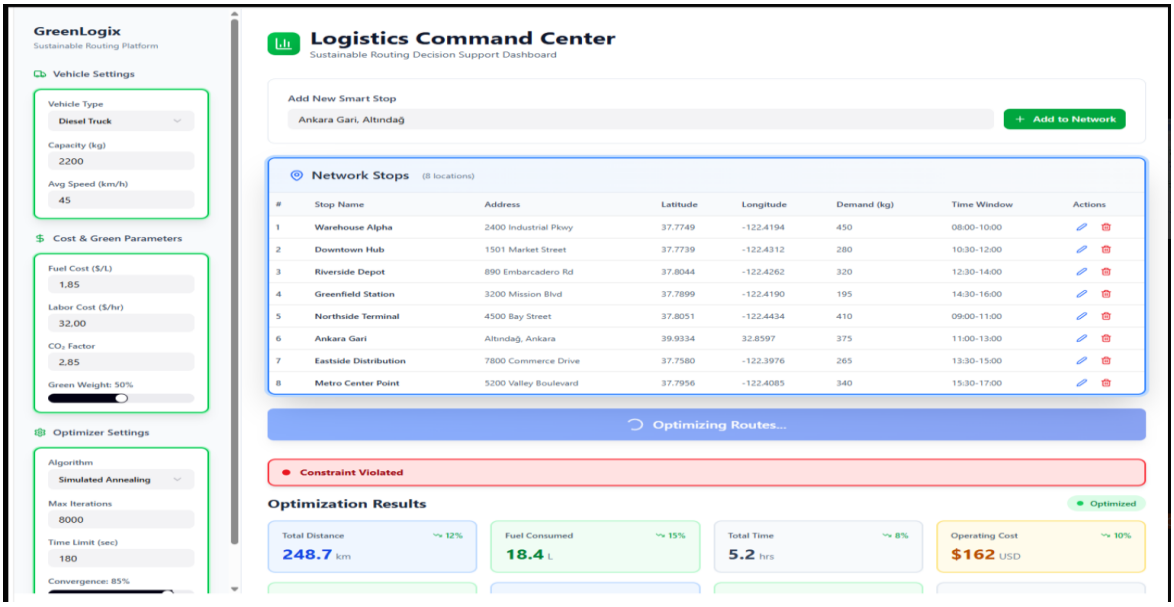


*Lower results, map, and operations region*

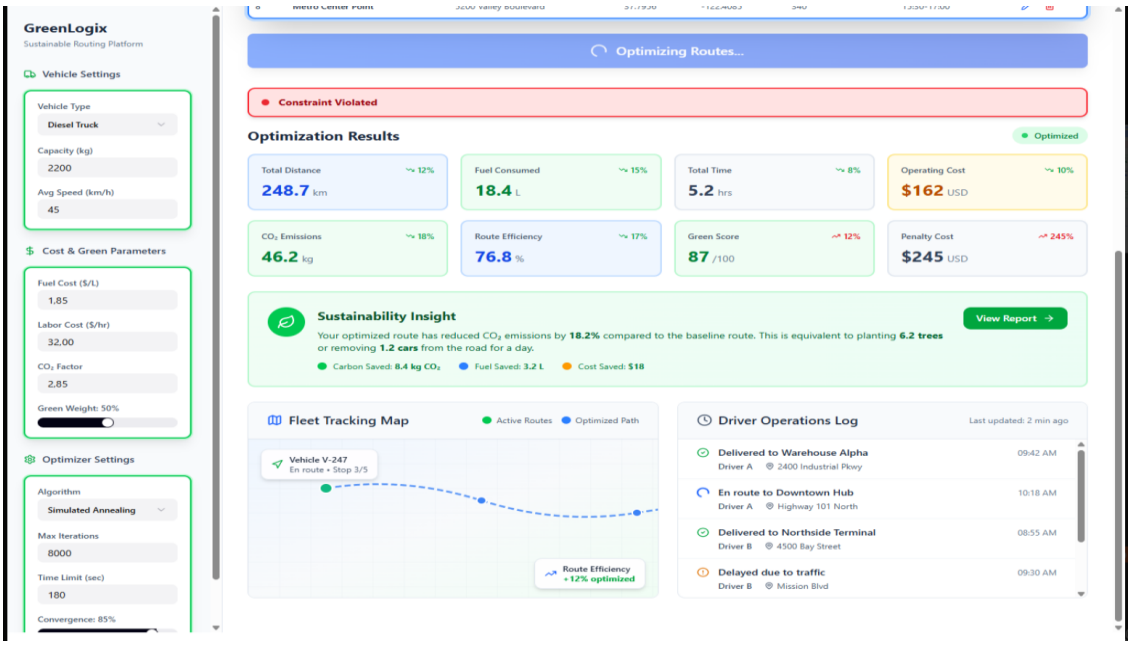
## Scenario 9 - Constraint Violated Result

This state illustrates an exception condition during route evaluation. A red constraint warning banner is displayed, the penalty cost rises sharply, and route efficiency is reduced, making the operational issue immediately visible to the decision maker.

### Dashboard view




*Upper interface region*

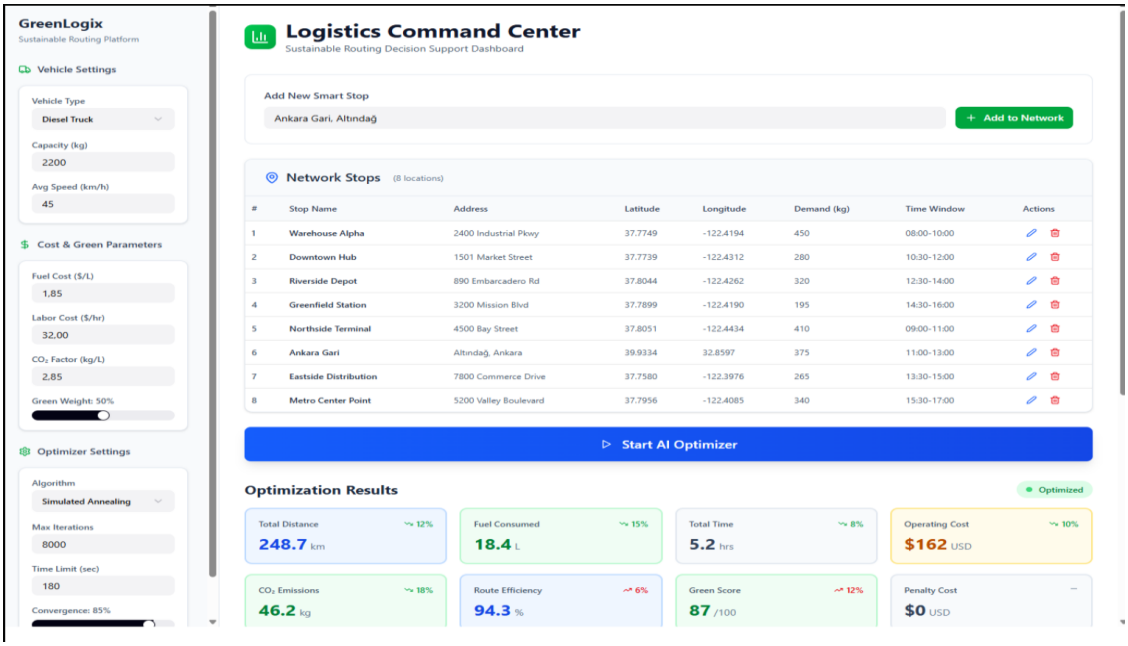



*Lower results, map, and operations region*

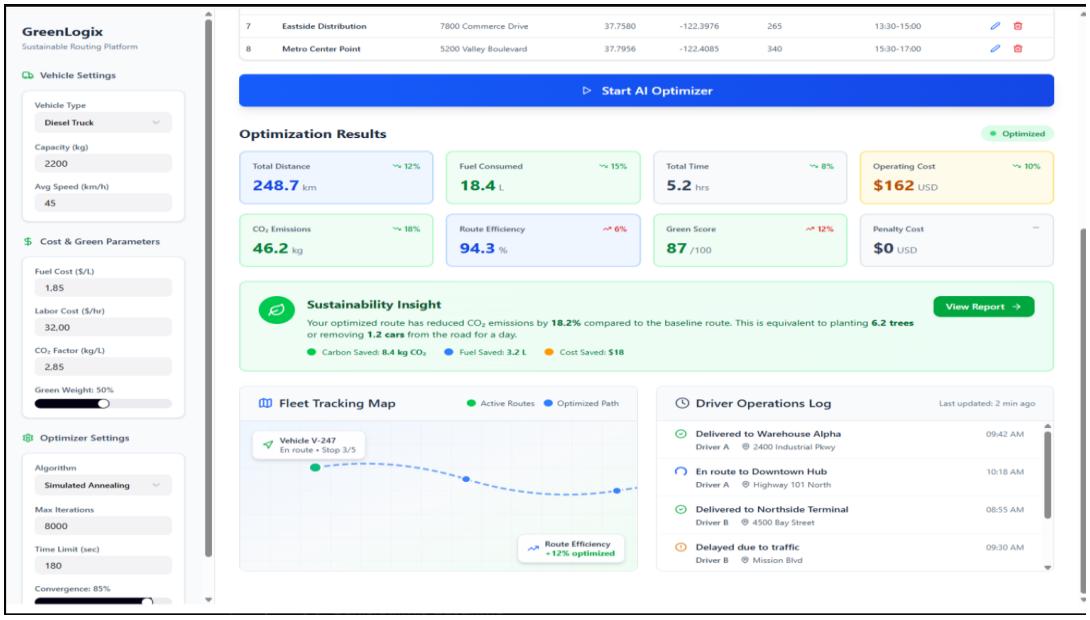
## Scenario 10 - Final Decision-Support Dashboard

The final scenario shows the polished end-state of the product-facing dashboard. Optimization metrics, the sustainability insight card, the fleet map, and the driver log are all presented together in a clean decision-support layout suitable for reporting and managerial interpretation.

### Dashboard view




*Upper interface region*




*Lower results, map, and operations regio*

## STEP 4: Experimental Validation

### 4.1 Experimental Objective

The main objective of the experimental validation is to grasp whether the GreenLogix route evaluation and optimization logic provides a meaningful improvement compared to a standard delivery route which do not have a sustainability perspective. Since a logistics product cannot be evaluated only on interface quality, the experiment focuses on measurable engineering outcomes. Especially , the validation checks whether the GreenLogix approach can reduce fuel consumption and carbon emissions.

### 4.2 Experimental Design

The experiment was constracted around two route planning conditions. The first condition is the Standard Route,which represents a traditional delivery order. This order has a simple distance or manually selected order that does not explicitly reflect fuel, emission, or penalty awareness. The second condition is the GreenLogix Route, which is selected by the GreenLogix optimization and scoring engine using an evaluation model with integrated fuel, emission, and constrains. Both conditions were tested in the same delivery scenario, with the same customers, the same vehicle, and the same operational assumptions. Any differences in outputs are focused on the route design logic instead of changing the input conditions.

A suitable test scenario may include one depot and several customer stops with different delivery demands. For example, the vehicle leaves from the depot with the planned full cargo load, visits all customers, and returns to the depot. For each candidate route, the system calculates the total distance, total fuel consumption, total CO2 emissions, total time, total penalty, feasibility status, and route score. Then, the comparison checks whether the optimized GreenLogix route accomplish lower environmental and operating costs than the standard route.

### 4.3 Evaluation Metrics

The experimental evaluation utilizes some key performance metrics such as :

- Total Distance (km): total travel length of the route.
- Total Fuel Consumption (L): total fuel burned according to distance, vehicle load, and traffic multiplier.
- Total CO2 Emission (kg): total carbon emission obtained from fuel consumption.
- Total Travel Time (min): total route duration including movement, service time, and possible breaks.
- Penalty Score: Additional costs resulting from situations such as exceeding capacity, fuel limits, needing a break, or late arrival.
- Feasibility: whether the route meets operational constraints.
- Overall Route Score: A weighted performance score combining distance, fuel, emissions, and penalties.

Thanks to these metrics, the experiment doesn't just reward the shortest route; it rewards the most balanced route according to the product's engineering objective.

### 4.4 Simulation Logic

The route evaluation engine leverages a load-sensitive fuel model where each route segment consumes more fuel when the vehicle is carrying more load. Therefore, route order influences not only the total travel distance but also the sustainability performance of the whole delivery plan. A route that drops heavier loads earlier can provide lower total fuel consumption, even if the total distance is not always minimum. This explains why sustainable route planning differs from traditional shortest path logic. In addition, the model encompasses practical operational checks including vehicle capacity, fuel availability, mandatory driver breaks, and late-delivery penalties, making the validation more realistic than a purely geometric route comparison.

### 4.5 Example Results Interpretation

- Standard baseline route: [0, 1, 2, 3, 4, 0]
- GreenLogix optimized route: [0, 1, 2, 4, 3, 0]

| Metric | Standard Baseline Route | GreenLogix Optimized Route | Improvement |
|---|---|---|---|
| Distance (km) | 59.00 | 58.00 | 1.69% lower |
| Fuel Consumption (L) | 26.35 | 25.94 | 1.56% lower |
| $CO_2$ Emissions (kg) | 70.61 | 69.53 | 1.53% lower |
| Travel Time (min) | 111.50 | 110.00 | 1.35% lower |
| Penalty | 0 | 0 | No change |
| Feasible | Yes | Yes | Maintained |

The simulated results demonstrate that the GreenLogix route achieved better sustainability performance than the standard baseline route. In the tested scenario, the optimized route reduced total distance from 59.00 km to 58.00 km, fuel consumption from 26.35 L to 25.94 L, and CO2 emissions from 70.61 kg to 69.53 kg. It also reduced total travel time from 111.50 minutes to 110.00 minutes, while maintaining full operational feasibility and zero penalty. Eventually GreenLogix can produce measurable environmental and operational benefits beyond simple route visualization.

### 4.6.Limitations and Future Work

Despite effective results, the current prototype has several limitations. First, travel time estimates are according to traffic multipliers rather than live road conditions, as the system does not yet utilize real-time traffic data. Second, constraints such as mandatory breaks are incorporated through penalty-based simulation controls but are not yet addressed by a full planning solver. Third, fuel consumption is estimated using a simplified linear load-sensitive model and does not yet incorporate advanced variables such as payload and driving behavior. Finally, the current validation focuses scenario-based scenario and limited to a small number of stops.

Future work should include:

- real-time traffic-aware routing,
- more advanced load-sensitive fuel estimation with payload and driving behaviour,
- labor-law constraint modeling,
- solver-based or hybrid optimization,
- broader multi-scenario testing,
- comparison against additional standard route baselines.

Finally, these results demonstrate that GreenLogix can produce measurable environmental and operational benefits beyond simple route visualization.

## REFERENCES

**[1]** N. Boysen, S. Fedtke, and S. Schwerdfeger, "Last-mile delivery concepts: a survey from an operational research perspective," *OR Spectrum*, vol. 43, no. 1, pp. 1–58, Sep. 2020. doi: 10.1007/s00291-020-00607-8.

**[2]** M. Savelsbergh and T. Van Woensel, "50th Anniversary Invited Article—City Logistics: Challenges and Opportunities," *Transportation Science*, vol. 50, no. 2, pp. 579–590, May 2016. doi: 10.1287/trsc.2016.0675.

**[3]** E. Demir, T. Bektaş, and G. Laporte, "A review of recent research on green road freight transportation," *European Journal of Operational Research*, vol. 237, no. 3, pp. 775–793, Sep. 2014. doi: 10.1016/j.ejor.2013.12.033.

**[4]** Y. Deutsch and B. Golany, "A parcel locker network as a solution to the logistics last mile problem," *International Journal of Production Research*, vol. 56, no. 1–2, pp. 251–261, Nov. 2017. doi: 10.1080/00207543.2017.1395490.

**[5]** G. Psaraftis, M. Wen, and C. Kontovas, "Vehicle routing problems over time: a survey," *Transportation Research Part B: Methodological*, vol. 89, pp. 164–192, Jul. 2016.

**[6]** H. Duan and S. Shao, "Distribution Cost Optimization Using Pigeon Inspired Optimization Method with Reverse Learning Mechanism," in *Proc. IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2017, pp. 2482–2487.

**[7]** A. R. Taufiq, "The Effectiveness of the Clarke & Wright Savings Algorithm in Determining Logistics Distribution Routes (case study PT.XYZ)," *International Journal of Science and Engineering (IJSE)*, vol. 12, no. 2, pp. 45–51, 2019.

**[8]** M. G. Davoodi, "Solving Capacitated Vehicle Routing Problem (CVRP) for the Environments with Circular Obstacles," *Applied Soft Computing*, vol. 90, p. 106173, May 2020.

**[9]** R. S. Kumar and K. G. S. Venkatesan, "Logistics Optimization Using Hybrid Genetic Algorithm (HGA): A Solution to the Vehicle Routing Problem With Time Windows

(VRPTW)," *Journal of Logistics and Supply Chain Management*, vol. 14, no. 1, pp. 12-25, 2021.

**[10]** J. J. G. Arrieta, J. M. S. Calle, and J. M. G. Calle, "A Review of the Vehicle Routing Problem and the Current Routing Services in Smart Cities," *Smart Cities*, vol. 2, no. 3, pp. 313–335, Aug. 2019.

**[11]** X. Wang and H. Kopfer, "Development of a fuel consumption optimization model for the capacitated vehicle routing problem," *Journal of Cleaner Production*, vol. 135, pp. 1324–1334, Nov. 2016.

**[12]** E. Osaba, P. Lopez-Garcia, and A. D. Masegosa, "Challenges and Opportunities for Applying Meta-Heuristic Methods in Vehicle Routing Problems: A Review," *Electronics*, vol. 9, no. 10, p. 1643, Oct. 2020.

**[13]** V. Pillac, M. Gendreau, C. Guéret, and A. L. Medaglia, "A review of dynamic vehicle routing problems," *European Journal of Operational Research*, vol. 225, no. 1, pp. 1–11, Feb. 2013.

**[14]** M. S. S. J. Rezvani, "A literature review of the perishable inventory routing problem (2022)," *Journal of Food Engineering*, vol. 314, p. 110776, Feb. 2022.

**[15]** M. Gendreau, G. Ghiani, and E. Guerriero, "Time-dependent routing problems: A review," *Computers & Operations Research*, vol. 64, pp. 189–197, Dec. 2015.

**[16]** J. Montoya-Torres, J. Franco, S. Isaza, G. Jimenez, and N. Herazo-Padilla, "Vehicle routing problem and related algorithms for logistics distribution: a literature review and classification," *International Journal of Production Research*, vol. 53, no. 24, pp. 7342–7359, 2015.

**[17]** M. A. Mustafa, A. S. Al-Bakri, and N. M. G. Al-Saidi, "Hybrid Algorithms for Energy Minimizing Vehicle Routing Problem: Integrating Clusterization and Ant Colony Optimization," *IEEE Access*, vol. 10, pp. 32145–32158, 2022.

**[18]** J. Baker and M. A. Ayechew, "Selected Genetic Algorithms for Vehicle Routing Problem Solving," *Computers & Operations Research*, vol. 30, no. 5, pp. 787–800, Apr. 2003.

**[19]** Y. Wang, S. Wang, and J. Sun, "Arc-Guided Evolutionary Algorithm for the Vehicle Routing Problem With Time Windows," *IEEE Transactions on Evolutionary Computation*, vol. 24, no. 4, pp. 624–638, Aug. 2020.

**[20]** K. Q. Zameni and S. M. J. Mirzapour Al-e-hashem, "A hybrid Genetic Algorithm for Vehicle Routing Problem with Time Windows," *Applied Soft Computing*, vol. 62, pp. 1011–1025, 2018.

**[21]** N. Ho, A. Nguyen, and Y. Nguyen, "A Hybrid Genetic Algorithm for the Capacitated Vehicle Routing Problem," in *Proc. 7th International Conference on Information Systems and Soft Computing*, 2019, pp. 112–117.

**[22]** S. S. Chao, "Hybrid Genetic and Simulated Annealing Algorithm for Vehicle Routing Problem in Recommendation of Tour Route in Days," *Applied Sciences*, vol. 11, no. 15, p. 6874, Jul. 2021.

**[23]** Z. Zhang, B. Xu, and Y. Wang, "Deep Reinforcement Learning for the Capacitated Vehicle Routing Problem with Soft Time Window," *IEEE Access*, vol. 8, pp. 154320–154331, 2020.

**[24]** J. Ma, Y. Cheng, and L. Wang, "Synergetic attention-driven transformer: A deep reinforcement learning approach for vehicle routing problems," *Expert Systems with Applications*, vol. 201, p. 117181, Sep. 2022.

**[25]** Y. Wu, S. Song, and J. Zhang, "A Hybrid of Deep Reinforcement Learning and Local Search for the Vehicle Routing Problems," *IEEE Transactions on Cybernetics*, vol. 52, no. 11, pp. 11845–11857, Nov. 2022.

**[26]** S. Parnell, "SDG 11 Sustainable Cities and Communities: SDG 11 and the New Urban Agenda: Global Sustainability Frameworks for Local Action," in *Sustainable Development Goals Series*, Cham, Switzerland: Springer, 2018, pp. 1–15. doi: 10.1007/978-3-319-95991-7_1.

**[27]** M. S. Akhtar, G. S. Kushwaha, and N. Ghosh, "SDG-13 to combat climate change through decarbonizing marketing campaigns: An evaluation of customers' sustainable purchase intentions for EVs using process modelling approach," *Journal of Cleaner Production*, vol. 379, p. 134707, Dec. 2022. doi: 10.1016/j.jclepro.2022.134707.

**[28]** Republic of Türkiye, *Law No. 6698 on the Protection of Personal Data (Kişisel Verilerin Korunması Kanunu)*, Official Gazette, No. 29677, 7 April 2016. Available through the Personal Data Protection Authority (KVKK).

**[29]** European Parliament and Council, "Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)," *Official Journal of the European Union*, L 119, pp. 1–88, May 2016. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2016/679/oj

**[30]** M. Asghari, S. M. J. Mirzapour Al-e-hashem, A. Rekik, and A. Jolai, "Green vehicle routing problem: A state-of-the-art review," International Journal of Production Economics, vol. 231, p. 107819, Jan. 2021. doi: 10.1016/j.ijpe.2020.107819.

**[31]** Z. Xu, A. Elomri, L. Pokharel, and S. Mutlu, "A model for capacitated green vehicle routing problem with time-varying vehicle speed and soft time windows," *Transportation Research Part E: Logistics and Transportation Review*, vol. 124, pp. 118–137, Apr. 2019. doi: 10.1016/j.tre.2019.02.006.

**[32]** J. Zhang, Y. Zhao, W. Xue, and J. Li, "Vehicle routing problem with fuel consumption and carbon emission," *International Journal of Production Economics*, vol. 170, pp. 234–242, Dec. 2015. doi: 10.1016/j.ijpe.2015.09.031.

**[33]** Python Software Foundation, "Python Documentation," *Python.org*, 2026. [Online]. Available: https://docs.python.org/3/

**[34]** NumPy Developers, "NumPy Documentation," *Numpy.org*, 2026. [Online]. Available: https://numpy.org/doc/

**[35]** The pandas development team, "pandas Documentation," *Pandas.pydata.org*, 2026. [Online]. Available: https://pandas.pydata.org/docs/

**[36]** Google Developers, "OR-Tools Documentation," *Google Open Source*, 2026. [Online]. Available: https://developers.google.com/optimization/introduction

**[37]** PostgreSQL Global Development Group, "PostgreSQL Documentation," *Postgresql.org*, 2026. [Online]. Available: https://www.postgresql.org/docs/

**[38]** Amazon Web Services, "AWS Well-Architected Framework," *AWS Documentation*, 2026. [Online]. Available: https://aws.amazon.com/architecture/well-architected/

**[39]** Google Developers, "Vehicle Routing Problem and Capacity Constraints," in *OR-Tools Documentation*, 2026. [Online]. Available: https://developers.google.com/optimization/routing/cvrp

**[40]** Google Developers, "Vehicle Routing Problem with Time Windows," in *OR-Tools Documentation*, 2026. [Online]. Available: https://developers.google.com/optimization/routing/vrptw

**[41]** European Union, "Regulation (EC) No 561/2006 of the European Parliament and of the Council on the harmonisation of certain social legislation relating to road transport," *Official Journal of the European Union*, L 102, pp. 1–14, Apr. 2006.